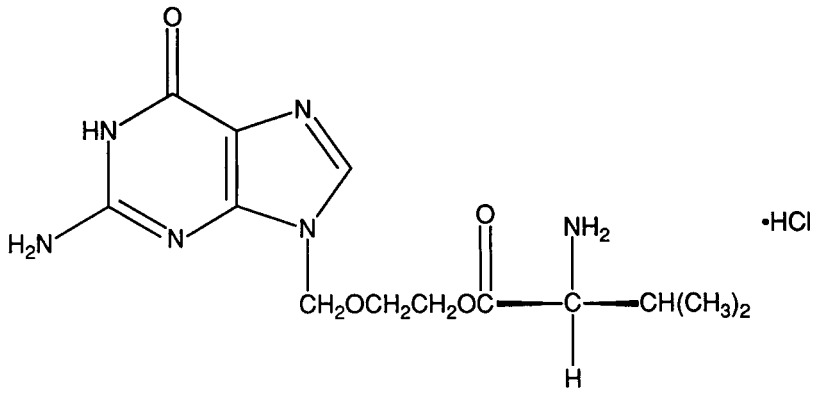
Pharmacokinetics of the acyclovir pro-drug valaciclovir after escalating single- and multiple-dose administration to normal volunteers

The pharmacokinetics and safety of the L-valyl ester pro-drug of acyclovir, valaciclovir (256U87), were investigated in two phase I, placebo-controlled trials in normal volunteers. These included a single-dose study with doses from 100 to 1000 mg (single cohort) and a multiple-dose investigation with doses from 250 to 2000 mg (five separate cohorts). In each cohort, eight subjects received valaciclovir and four subjects received placebo. Pharmacokinetic findings for valaciclovir and acyclovir were consistent in the two studies. Valaciclovir was rapidly and extensively converted to acyclovir, resulting in significantly greater acyclovir bioavailability (~ threefold to fivefold) compared with that historically observed with high-dose (800 mg) oral acyclovir. At the higher valaciclovir doses, acyclovir maximum concentration and daily area under the concentration-time curve approximated those obtained with intravenous acyclovir. The favorable safety profile and enhanced acyclovir bioavailability from valaciclovir administration has prompted additional clinical evaluations for zoster and herpes simplex virus treatment, as well as cytomegalovirus suppression in immunocompromised patients. (CLIN PHARMACOL THER 1993;54:595-605.)

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Acyclovir [9-(2-hydroxyethoxymethyl)guanine] is an acyclic purine nucleoside analog with in vitro and in vivo inhibitory activity against human herpesviruses, including herpes simplex type 1 (HSV-1) and type 2 (HSV-2), varicella-zoster virus (VZV), Epstein-Barr virus, and cytomegalovirus. The in vitro sensitivity of these viruses to acyclovir and the efficacy of acyclovir in the treatment and suppression of related infections has been extensively reviewed elsewhere.[1] Although acyclovir has shown efficacy in the treatment and suppression of HSV-1, HSV-2, and VZV, the usefulness of oral acyclovir against the less sensitive viruses is limited as a result of its low bioavailability. In humans, the average oral bioavailability of acyclovir after administration of a 200 mg dose is approximately 20%,[2] and this decreases to about 10% with 800 mg dosing (package insert for Zovirax capsules, tablets, and suspension. Burroughs Wellcome Co., Research Triangle Park, N.C.). Because of its low oral bioavailability, attainment of plasma concentrations adequate for inhibition of the less sensitive viruses is difficult, and intravenous administration may be necessary for improved efficacy. For this reason, the pro-drug approach has been investigated at Wellcome Research Laboratories for more than a decade to improve the bioavailability of acyclovir. Studies of two congeners of acyclovir[3,4] with enhanced acyclovir delivery were previously halted because chronic toxicity profiles in laboratory animals were less favorable than those for acyclovir itself. The toxicities were hypothesized to be the result of phosphorylation of unconverted pro-drug. Subsequent work to develop a pro-drug that would escape phosphorylation before conversion to acyclovir has led to selection of the L-valyl ester of acyclovir, valaciclovir (Valtrex, 256U87) (Structure), as the most promising compound

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Structure of valaciclovir, the L-valyl ester of acyclovir (2-[2-amino-1, 6-dihydro-6-oxo-9*H*-purin-9-yl)methoxy]ethyl-*L*-valinate hydrochloride). Acyclovir constitutes 69.4% of the molecular weight of valaciclovir base.

to date.[5] The findings from preclinical studies have indicated that, compared with oral acyclovir administration, valaciclovir administration to rats and monkeys yields 3 to 4 times greater acyclovir bioavailability.[6,7] Available findings from animal toxicology studies with valaciclovir are consistent with those for acyclovir.[8] Because of the enhanced acyclovir bioavailability and favorable toxicology profile, clinical studies were initiated to investigate valaciclovir in humans. We report here on the initial single-dose and multiple-dose phase I studies evaluating the safety, tolerance, and pharmacokinetics of valaciclovir in normal volunteers.

METHODS

Study designs. Both the single- and multiple-dose studies were randomized, double-blind placebo-controlled investigations of oral valaciclovir and were conducted at Harris Laboratories, Inc. (Lincoln, Neb.). Drug substance was supplied as 50, 200, and 250 mg capsules or matching placebo.

The single-dose study consisted of a single cohort of 12 healthy male volunteers (eight receiving active drug; four receiving placebo) who received valaciclovir at doses of 100, 250, 500, 750, and 1000 mg sequentially with a washout period of at least 7 days between treatments. The multiple-dose study consisted of five cohorts, each with 12 healthy male volunteers (eight receiving active drug; four receiving matching placebo), who received valaciclovir doses of 250, 500, 1000, 1500, or 2000 mg. Doses were administered every 5 hours, four times a day for 11 days with the following exceptions: the second dose was not administered on study day 1 (to aid in pharmacokinetic analysis after the first dose), and only the morning dose was administered on study day 11.

Subjects. Subjects for both studies were community volunteers at risk of developing primary or recurrent HSV infections and included men from 18 to 40 years of age. Each subject was to have a body weight between 60 and 85 kg and within 10% of ideal body weight. Subjects were excluded from investigation if they had signs and symptoms of active HSV or VZV infection; renal or hepatic impairment; malabsorption syndrome, or other gastrointestinal dysfunction that might interfere with drug absorption; anemia or blood donation within 8 weeks of the trial; or history of allergy or intolerance to acyclovir. Subjects were also excluded if they had received any prescription or over-the-counter medication on a long-term basis or within 48 hours of administration of test medication, if they were known or suspected heavy users of alcohol or tobacco or of illicit drugs, or if they had participated in any other investigational drug trial within 8 weeks. All subjects included in the studies gave informed consent for participation before enrollment.

Safety assessments. Laboratory and clinical evaluations were conducted to assess safety. Within 14 days before test medication was administered, a physical examination was performed for each volunteer, and medical history and demographic data were recorded. The physical examination included a review of systems and oral temperature, as well as sitting pulse rate, respiratory rate, and blood pressure. Baseline laboratory studies performed before study drug administration included hematologies consisting of hemoglobin, hematocrit, red blood cell count, white blood cell count with differential, and platelet count; blood chemistries consisting of urea nitrogen, creatinine, total bilirubin, alanine transaminase, aspartate transaminase, lactic dehydrogenase, and γ-glutamyl trans-

ferase; and urinalysis with microscopic analysis. A urine drug screen was also performed.

For the single-dose study, laboratory studies were repeated immediately before and 24 hours after each drug administration and again at 7 to 10 days after the last dose administration. For the multiple-dose study, blood for chemistry and hematology analysis and urine for urinalysis were collected before the first dose administration on day 1, on days 5 and 11, and 7 to 10 days after the last day of dosing.

Diet. For the single-dose investigation, subjects fasted from 8 hours before until 4 hours after receiving each dose of study medication. Each volunteer was fed a standardized meal at 4 and 10 hours after each dose. Alcoholic beverages were prohibited for 48 hours before and during each dosing period. Xanthine-containing beverages and foods were also prohibited during each dosing period. Water was permitted ad libitum.

For the multiple-dose study, subjects were fed a light snack at 10 PM each night of the study. Volunteers were fasted from 9 hours before until 5 hours after the first dose on days 1 and 11, when blood samples for complete pharmacokinetic profiles were obtained. Otherwise, subjects received regular, balanced meals at breakfast, lunch, and dinner while at the study site. The schedule of doses and meals provided a 4-hour fast before all other dose administrations.

Pharmacokinetic and statistical analyses. For the single-dose study, blood samples (5 ml) were obtained before dose administration, at 15, 30, and 45 minutes, and at 1, 1½, 2, 2½, 3, 4, 6, 8, 10, 12, 16, and 24 hours after each dose administration. In the multiple-dose study, blood samples were obtained after administration of the first dose on study days 1 and 11. On day 1, samples were collected before dose administration, at 15, 30, and 45 minutes, and at 1, 1½, 2, 3, 4, 5, 6, 8, and 10 hours after the first dose. On day 11, blood samples were collected at the same times but with additional samples at 12, 16, and 24 hours after dosing. In both studies, 24-hour urine collections were obtained for determination of recovery of valaciclovir, acyclovir, and the acyclovir metabolite 9-[(carboxymethoxy)methyl]guanine (CMMG). In the multiple-dose study, the 24-hour urine collections were obtained on days 3 and 10 of drug administration.

Pharmacokinetic analyses of plasma acyclovir concentration data were conducted by noncompartmental methods and included parameter estimation of peak concentration (C_{max}), time to peak concentration (t_{max}), area under the concentration-time curve (AUC), and apparent elimination half-life ($t_{½}$). C_{max} and t_{max} were obtained by direct inspection of the data. AUC was estimated by trapezoidal approximation up to the last measurable concentration (C_T) with extrapolation to infinite time by addition of the term C_T/λ, in which λ was estimated by regression of the log(concentration)-time data in the elimination phase. The $t_{½}$ was calculated as $\ln(2)/\lambda$. In the multiple-dose study, AUC for the last dose was corrected for residual drug attributable to previous doses by subtraction of C_0/λ, in which C_0 denotes the predose concentration.

The assessment of dose proportionality and other possible dose-related differences in pharmacokinetic parameter estimates was conducted by ANOVA with use of SAS software.[9] For dose proportionality evaluations, C_{max} and AUC were normalized for differences in valaciclovir dose before analysis. Repeated-measures ANOVA was conducted for data from the single-dose study in which subjects received each of five doses. In the event of a significant dose effect for a given parameter, post hoc tests were performed to identify specific differences.

Assay for valaciclovir in plasma. The levels of valaciclovir in plasma were determined by a 15-minute, reversed-phase HPLC assay with use of an HPLC system described previously.[10] A detailed description of the assay, including validation data, will be published elsewhere (Burnette TC, de Miranda P. Unpublished data). In brief, plasma samples were deproteinated and acidified by addition of trichloroacetic acid (TCA; final concentration of 3%) followed by centrifugation (9000*g*, for 10 minutes at 4° C). The supernatants were analyzed on an Adsorbosphere C_{18} column (4.6 mm × 25 cm; Alltech Associates/Applied Science, Avondale, Pa.) eluted with mobile phase (100 mmol/L ammonium formate buffer, pH 3.5, containing 10% acetonitrile) at a flow rate of 1 ml/min. The ultraviolet absorbance of the column effluent was monitored at 254 nm and 280 nm. The retention time for valaciclovir was approximately 8 minutes. The concentration of valaciclovir in each sample was calculated from a standard curve (log of the valaciclovir concentration versus log of the peak area at 254 nm) constructed from standards (0.25 to 10 μmol/L; 0.08 to 3.24 μg/ml) analyzed under identical conditions. The recovery of valaciclovir from spiked plasma controls by TCA precipitation was 103% to 107%. This assay had an overall precision of greater than 95% with a bias of −4% to −1%.

Assay for acyclovir in plasma. The determination of acyclovir concentrations in plasma was conducted

Table I. Demographics of volunteers in phase I single- and multiple-dose studies of valaciclovir

Demographic characteristic	*Single-dose study*		*Multiple-dose study*				
	Valaciclovir recipients	*Placebo recipients*	*Valaciclovir recipients*				
			250 mg	*500 mg*	*1000 mg*	*1500 mg*	*2000 mg*
Age (yr)	26.6 ± 7.3	27.2 ± 6.1	26.2 ± 7.7	25.9 ± 6.3	24.0 ± 3.3	30.1 ± 4.8	25.5 ± 6.2
Race							
White	6	4	6	8	6	8	8
Black	—	—	1	—	—	—	—
Other	2	—	1	—	2	—	—
Weight (kg)	74.6 ± 8.1	75.2 ± 11.3	70.5 ± 7.0	73.8 ± 7.7	69.5 ± 8.0	74.0 ± 7.1	74.0 ± 5.5
Height(cm)	176.2 ± 8.3	177.5 ± 9.1	176.7 ± 9.0	178.1 ± 7.1	176.5 ± 5.9	177.2 ± 6.3	178.9 ± 6.2
BSA (m^2)	1.91 ± 0.14	1.95 ± 0.19	1.87 ± 0.12	1.90 ± 0.11	1.84 ± 0.12	1.92 ± 0.13	1.92 ± 0.10
CL_{CR} (ml/min/1.73 m^2)	71.8 ± 7.1	72.0 ± 10.5	90.7 ± 11.6	79.7 ± 7.2	81.0 ± 8.9	73.8 ± 7.4	79.8 ± 9.6

Data are mean values ± SD.
BSA, Body surface area; CL_{CR}, creatinine clearance.

by scintillation proximity radioimmunoassay as reported elsewhere.[11,12] The lower limit of quantitation of the assay as conducted on samples obtained in the studies was 0.0023 μg/ml (0.01 μmol/L).

Assay for valaciclovir, acyclovir, and CMMG in urine. As for valaciclovir in plasma, the levels of valaciclovir, acyclovir, and CMMG in urine were also determined by HPLC but with a different method. Urine samples, acidified with 1% TCA at the time of collection, were filtered through Centrifree micropartition units (Amicon Corporation, Danvers, Mass.) under centrifugation (2000*g* for 10 to 20 minutes at 4° C). The urine ultrafiltrates along with the aqueous standards were analyzed on an Adsorbosphere C_{18} column (Alltech Associates/Applied Science) eluted by a gradient generated with 50 mmol/L ammonium formate buffer (pH 3.5) and 50 mmol/L ammonium phosphate buffer (pH 7.2) containing 50% acetonitrile at a flow rate of 1 ml/min. The approximate retention times for CMMG, acyclovir, and valaciclovir were 14, 20, and 27 minutes, respectively. The identity of each analyte in the urine was confirmed by comparison of its 254 to 280 nm peak area ratio to that of the authentic compound in aqueous solution. The concentration of each analyte was calculated as described above for valaciclovir in plasma. Concentration ranges for standards of 0.5 to 50 μmol/L (0.16 to 16.2 μg/ml) of valaciclovir, 1 to 200 μmol/L (0.23 to 45.0 μg/ml) of acyclovir, and 1 to 50 μmol/L (0.24 to 12.0 μg/ml) of CMMG were found suitable for analyses. Urine ultrafiltrates were diluted with sterile, filtered deionized water as needed for each analyte.

RESULTS

Subjects. A summary of the demographic characteristics of the subjects enrolled in the studies is presented in Table I. In both the single-dose study (single cohort) and the multiple-dose study (separate cohorts), recipients of valaciclovir and placebo were well matched with respect to age, body weight, height, and estimated creatinine clearance. The latter was calculated according to the method of Mawer et al.[13] Similarly, demographic characteristics at entry for drug recipients in the different cohorts and across studies were generally comparable.

Not all subjects enrolled completed all doses. In the single dose investigation, one subject randomized to valaciclovir withdrew for personal reasons after completing only the initial 100 mg dose period. In the multiple-dose study, one subject receiving 500 mg valaciclovir withdrew on day 6 and another valaciclovir recipient (2000 mg) withdrew on day 7, both because of personal reasons unrelated to study participation.

Single-dose pharmacokinetics. A plot of the average plasma acyclovir concentrations obtained from subjects who received valaciclovir in the single-dose study is presented in Fig. 1. Acyclovir pharmacokinetic parameter estimates are presented in Table II. Mean (±SD) acyclovir peak concentrations ranged from 0.83 ± 0.14 to 5.65 ± 2.37 μg/ml after valaciclovir doses of 100 and 1000 mg, respectively. Corresponding values for AUC were 2.28 ± 0.40 and 19.52 ± 6.04 hr · μg/ml, respectively. The findings from the statistical analysis of both dose-normalized C_{max} and AUC indicated a less than proportional increase with increasing dose of valaciclovir, although

Multiple-dose study				
Placebo recipients				
250 mg	*500 mg*	*1000 mg*	*1500 mg*	*2000 mg*
25.8 ± 7.6	24.0 ± 8.5	23.4 ± 2.2	24.0 ± 5.7	24.6 ± 8.5
1	4	4	4	3
3	—	—	—	—
—	—	—	—	1
75.6 ± 7.2	72.5 ± 8.3	77.5 ± 7.7	70.2 ± 9.7	75.4 ± 4.9
179.1 ± 5.4	180.0 ± 7.4	188.0 ± 5.8	178.4 ± 3.0	176.6 ± 6.2
1.95 ± 0.13	1.90 ± 0.14	1.97 ± 0.10	1.87 ± 0.15	1.95 ± 0.06
80.7 ± 11.3	85.2 ± 10.9	85.8 ± 6.5	77.8 ± 8.5	81.5 ± 13.7

the deviation from linearity was less pronounced and occurred at a higher dose for AUC than for C_{max}. The time of occurrence of peak acyclovir concentration also displayed dose dependency, with greater values as valaciclovir doses increased. Acyclovir t_{max} approximately doubled from an average of 0.88 ± 0.13 to 1.75 ± 0.63 hours over the 100 to 1000 mg dose range of valaciclovir. The overall mean $t_{1/2}$ for acyclovir was 2.86 ± 0.39 hours and did not differ with dose.

Plasma concentrations of valaciclovir were ≤0.5 μg/ml at all sampling times in the single-dose study and were typically below the lower limit of quantitation of the assay (0.08 μg/ml) within 3 hours after administration. With the exception of peak concentrations, parameter estimates for valaciclovir were generally unobtainable. Peak concentrations of valaciclovir averaged approximately 4% (on a molar basis) of the corresponding C_{max} for acyclovir; no differences in this proportion were observed across the different valaciclovir dose levels.

Multiple-dose pharmacokinetics. Mean concentration-time profiles for acyclovir from the multiple-dose investigation are displayed in Fig. 2, and pharmacokinetic parameter estimates are presented in Table III. Peak concentrations after the first dose on day 11 ranged from 2.11 ± 0.33 to 8.49 ± 1.67 μg/ml after valaciclovir doses of 250 and 2000 mg, respectively. Mean acyclovir AUC values ranged from 5.59 ± 1.04 to 29.50 ± 6.36 hr · μg/ml over the same dose range. As observed in the single-dose investigation, there was a less than linear increase in both C_{max} and AUC with increasing doses of valaciclovir and a corresponding increase in acyclovir t_{max}. Overall average acyclovir $t_{1/2}$ values in the multiple-dose study were 2.65 ± 0.33 hours after the first dose on day 1 and 3.13 ± 0.54 hours on day 11. Although no dose-related changes were observed in this parameter on either day 1 or day 11, $t_{1/2}$ estimates on day 11 at the higher valaciclovir doses were significantly longer than on day 1 (paired t test, $p < 0.05$). These differences are probably attributable to the longer sampling intervals used on day 11 rather than a result of changes in acyclovir disposition. At a given dose level of valaciclovir common to both the single- and multiple-dose studies, with the single exception of $t_{1/2}$ on day 1 after 1000 mg administration, no significant differences were observed in any parameter estimates between studies (two-sided t test, $p \geq 0.05$).

The highest valaciclovir concentration observed in the multiple-dose study was 0.87 μg/ml at 1½ hours after the first 2000 mg dose in one subject; concentrations at ≥3 hours after dosing were below the lower limit of quantitation. Valaciclovir C_{max} for all other volunteers was ≤0.56 μg/ml. As in the single-dose study, valaciclovir was typically nonquantifiable within 3 hours after administration and pharmacokinetic parameters other than peak concentration were not estimable. Also in agreement with findings from the single-dose investigation, the molar ratio of peak concentrations of valaciclovir to acyclovir averaged 4% at the different valaciclovir dose levels; this did not differ during the 11-day dosing duration. No accumulation in valaciclovir plasma concentrations was observed with the dosing regimens used in this investigation.

Urinary recovery. Urinary recovery data from the single- and multiple-dose studies are presented in Tables IV and V, respectively. For all doses, after both single- and multiple-dose administration, less than 1%

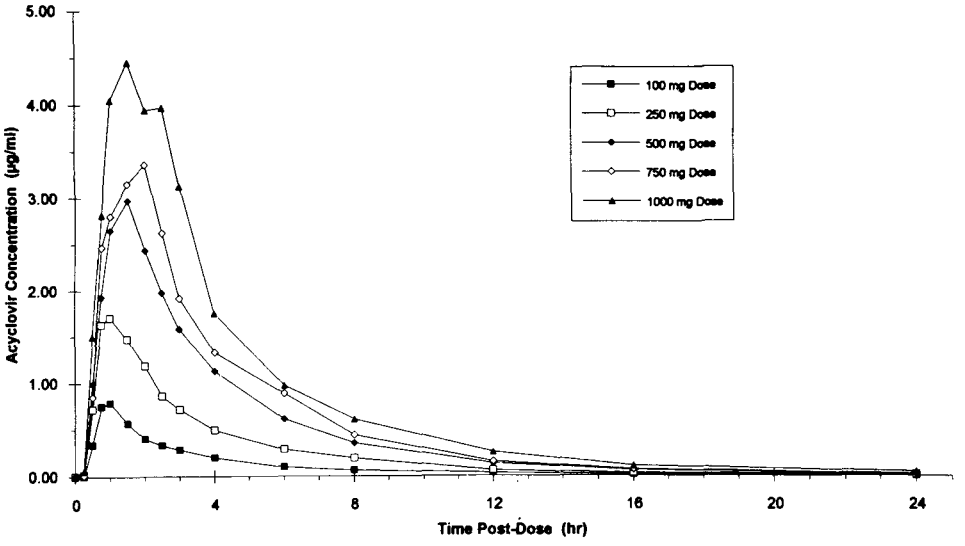


Fig. 1. Mean plasma acyclovir concentrations after single-dose administration of valaciclovir to normal volunteers. A single cohort of subjects received valaciclovir at each dose level ($n = 7$ except for the 100 mg dose, for which $n = 8$).

Table II. Acyclovir pharmacokinetic parameter estimates after single-dose administration of valaciclovir to normal volunteers

Valaciclovir dose (mg)	C_{max} *(µg/ml)*	t_{max} *(hr)*	*AUC (hr · µg/ml)*	$t_{1/2}$ *(hr)*
100 ($n = 8$)	0.83 ± 0.14[a]	0.88 ± 0.13[a]	2.28 ± 0.40[a]	2.76 ± 0.55
250 ($n = 7$)	2.15 ± 0.50[a]	1.04 ± 0.34[a,b]	5.76 ± 0.60[a]	2.78 ± 0.36
500 ($n = 7$)	3.28 ± 0.83[b]	1.50 ± 0.58[b,c]	11.59 ± 1.79[a]	2.88 ± 0.18
750 ($n = 7$)	4.17 ± 1.14[b]	1.46 ± 0.55[a,b,c]	14.11 ± 3.54[b]	2.88 ± 0.49
1000 ($n = 7$)	5.65 ± 2.37[b]	1.75 ± 0.63[c]	19.52 ± 6.04[a,b]	3.03 ± 0.13

Data are mean values ± SD.
C_{max}, Maximum plasma concentration; t_{max}, time to reach C_{max}; AUC, area under the plasma concentration–time curve; $t_{1/2}$, half-life.
For a given parameter, estimates with common letters are not significantly different (i.e., $p > 0.05$) by repeated-measures ANOVA. (C_{max} and AUC were dose-normalized before analysis.)
No dose-related differences were observed in $t_{1/2}$.

of the valaciclovir dose was recovered unchanged in the 24-hour urine collections. For acyclovir in the single-dose study, the percentage recovered declined with increasing dose and ranged from an average of 76% at the 100 mg dose to 44% at the 1000 mg dose. However, this decrease was not observed in the multiple-dose study, and the percentage recovered as acyclovir remained relatively constant from about 40% to 50% across dose levels. Recovery of the acyclovir metabolite, CMMG, was consistent across studies and dose levels, ranging from approximately 7% to 12% of the dose. In both studies and for all doses, acyclovir consistently accounted for 80% to 85% of the total urinary recovery. Reasons for the difference in dose-related patterns of acyclovir recovery observed between the two studies are unknown. However, because the different valaciclovir doses in the single-dose study were received by the same cohort of volunteers, it may be speculated that the pattern of dose-related decrease in acyclovir excretion observed in this study is more realistic than that observed for the multiple-dose investigation in which different doses were received by different subjects.

Safety assessments. In the single-dose study, a total of 15 mild to moderate adverse events (eight in the valaciclovir group and seven in the placebo group) were experienced by five volunteers (three in the valaciclovir group and two in the placebo group). There were no adverse events rated as severe. Eleven of the 15 events were deemed by the investigator to be "not as-

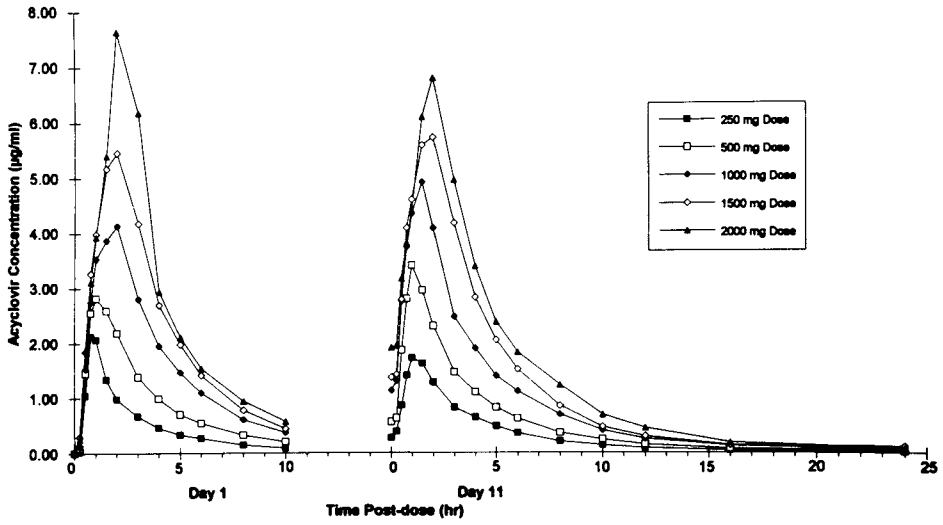


Fig. 2. Mean plasma acyclovir concentrations after multiple-dose administration of valaciclovir to normal volunteers. Each dose level of valaciclovir was administered to a separate cohort (n = 8 except for the 500 and 2000 mg doses on day 11, for which n = 7). Doses were administered every 5 hours, four times daily.

sociated with study drug" and included headache, asthenia, diarrhea, and somnolence. The only adverse events associated with study medication were gastrointestinal in nature and included abdominal pain and change in stool color (two occasions) in one placebo recipient and a change in stool color in one valaciclovir recipient after the 250 mg dose. Of all reported adverse events, no evidence of a relationship with dose was observed. No clinically significant changes in laboratory parameters were noted with either treatment group. Increasing anemia developed in one volunteer who received valaciclovir during the study, but subsequent review of prestudy medical records indicated that he had a previously undiagnosed iron-deficiency anemia.

In the multiple-dose investigation, a total of 105 adverse events (75 in the valaciclovir group and 30 in the placebo group) were reported by 43 of the 60 volunteers. These are summarized in Table VI. Eighteen of the adverse events in the valaciclovir group were considered to be associated with test medication and included multiple headaches in four volunteers at the 500 mg dose and in three volunteers at the 2000 mg dose. Two subjects receiving the 2000 mg dose experienced multiple episodes of diarrhea and two in the same group experienced nausea. Other adverse experiences that were considered to be associated with test medication were pain, leg cramps, dyspepsia, dizziness, and photophobia. Two placebo recipients reported headaches that the investigator judged to be related to test medication. Adverse events not associated with study medication included headache, rhinitis, pharyngitis, pruritus, abdominal pain, back pain, nausea, dizziness, asthenia, somnolence, cough, infection, arrhythmia, tachycardia, vasodilation, flatulence, thirst, vomiting, arthralgia, myalgia, hyperkinesia, nervousness, voice alteration, rash, ear disorder, lacrimation disorder, ear pain, and eye pain in the valaciclovir recipients, as well as headache, rhinitis, pharyngitis, abdominal pain, nausea, dizziness, cough, hypertension, fever, abdominal pain, hypertension, dry mouth, nausea, tooth disorder, dizziness, cough, pharyngitis, rhinitis, rash, sweating, and dry eyes in placebo recipients. Few changes in laboratory parameters were observed. Two subjects who received 2000 mg valaciclovir had total white blood cell counts below normal at posttreatment follow-up ($2.62 \times 10^3/mm^3$ and $3.81 \times 10^3/mm^3$). All white blood cell counts for these subjects had been within normal limits while receiving test medication. One placebo and five valaciclovir recipients had mild elevations in serum transaminase levels that were deemed not to be clinically significant.

DISCUSSION

Consistent with preclinical investigations, the findings from these phase I studies in normal human volunteers have shown that valaciclovir is well absorbed

Table III. Acyclovir pharmacokinetic parameter estimates after multiple-dose administration of valaciclovir to normal volunteers

Valaciclovir dose (mg)	C_{max} *(μg/ml)*		t_{max} *(hr)*		*AUC (hr · μg/ml)*	
	Day 1	*Day 11*	*Day 1*	*Day 11*	*Day 1*	*Day 11*
250	2.24 ± 0.28[a]	2.11 ± 0.33[a]	0.84 ± 0.13[a]	1.25 ± 0.44[a]*	5.59 ± 1.04[a]	5.66 ± 1.09[a]
500	3.45 ± 1.10[b]	3.69 ± 0.87[a]	1.28 ± 0.49[a]	1.21 ± 0.57[a]	10.66 ± 1.70[a,b]	9.88 ± 2.01[a]
1000	4.80 ± 1.48[c]	4.96 ± 0.64[b]	1.88 ± 0.58[b]	1.56 ± 0.32[a,b]	18.30 ± 2.94[b,c]	15.70 ± 2.27[b]*
1500	6.19 ± 1.63[c]	6.39 ± 1.70[b]	1.81 ± 0.59[b]	1.88 ± 0.58[b,c]	24.04 ± 5.80[c]	21.37 ± 5.31[b]
2000	8.30 ± 1.43[c]	8.49 ± 1.67[b]	2.13 ± 0.58[b]	2.29 ± 0.91[c]	29.50 ± 6.36[c]	27.37 ± 7.94[b]

Data are mean values ± SD.
n = 8 except for 500 and 2000 mg doses on day 11 (n = 7).
*Denotes significant difference from day 1 (paired t test; $p < 0.05$).
For a given parameter, estimates with common letters are not significantly different (i.e., $p > 0.05$) by post hoc t test from ANOVA. (C_{max} and AUC were dose-normalized before analysis.)
No dose-related differences were observed in $t_{1/2}$.

Table IV. Urinary recovery (as molar percentage of dose) of valaciclovir and metabolites after single-dose administration

Valaciclovir dose (mg)	*Valaciclovir (%)*	*Acyclovir (%)*	*CMMG (%)*	*Total (%)*
100 (n = 8)	0.79 ± 0.28	75.9 ± 8.6[a]	11.9 ± 3.7[a]	88.6 ± 11.4[a]
250 (n = 7)	0.84 ± 0.19	65.0 ± 8.9[a]	12.2 ± 3.2[a]	78.0 ± 10.5[a]
500 (n = 7)	0.78 ± 0.21	71.6 ± 6.1[a]	12.4 ± 2.9[a]	84.8 ± 8.6[a]
750 (n = 7)	0.53 ± 0.10	49.0 ± 11.8[b]	10.0 ± 3.1[b]	59.5 ± 13.0[b]
1000 (n = 7)	0.53 ± 0.11	44.3 ± 10.0[b]	8.7 ± 3.5[b]	53.5 ± 14.9[b]

Data are mean ± SD molar percentage of dose.
CMMG, 9-[(Carbodymethoxy)methyl]guanine.
For a given compound, estimates with common letters are not significantly different (i.e., $p > 0.05$) by repeated-measures ANOVA.
No dose-related differences were observed in valaciclovir recovery.

Table V. Urinary recovery (as molar percentage of dose) of valaciclovir and metabolites after multiple-dose administration

Valaciclovir dose (mg)	*Valaciclovir*		*Acyclovir*		*CMMG*		*Total*	
	Day 3	*Day 10*	*Day 3*	*Day 10*	*Day 3*	*Day 10*	*Day 3*	*Day 10*
250	0.45 ± 0.17[a]	0.43 ± 0.19[a]	54.1 ± 10.8	49.2 ± 8.3	8.8 ± 1.9	8.6 ± 1.9	63.4 ± 12.4	58.2 ± 9.8
500	0.53 ± 0.28[a,b]	0.50 ± 0.23[a,b]	41.3 ± 16.2	47.4 ± 14.2	8.5 ± 4.6	10.3 ± 4.1	50.3 ± 20.9	58.2 ± 18.2
1000	0.55 ± 0.16[a,b]	0.54 ± 0.13[a,b]	43.2 ± 8.0	36.7 ± 6.2	8.3 ± 1.6	7.7 ± 2.0	52.0 ± 9.3	44.9 ± 8.1*
1500	0.69 ± 0.16[b,c]	0.66 ± 0.20[b,c]	44.4 ± 10.9	42.0 ± 14.1	9.4 ± 2.2	9.9 ± 3.8	54.4 ± 12.9	52.5 ± 17.6
2000	0.83 ± 0.13[c]	0.82 ± 0.18[c]	43.6 ± 5.1	41.9 ± 7.7	9.6 ± 1.2	9.6 ± 2.0	54.0 ± 6.0	52.3 ± 9.7

Data are mean ± SD molar percentage of the dose.
n = 8 except for 500 and 2000 mg doses on day 11 (n = 7).
*Denotes significant difference from day 3 (paired t test; $p < 0.05$).
For a given compound, estimates with common letters are not significantly different (i.e., $p > 0.05$) by post hoc t test from ANOVA.
No dose-related differences were observed in acyclovir, CMMG, or total recovery.

and is rapidly and extensively converted to acyclovir, as well as that the bioavailability of acyclovir from administration of valaciclovir is significantly enhanced relative to that from administration of oral acyclovir itself. Although the specific mechanism of valaciclovir conversion has not yet been completely identified, preclinical and in vitro studies have indicated that acyclovir formation results from first-pass intestinal and hepatic metabolism by an enzymatic hydrolysis.[5,6]

The enhanced acyclovir bioavailability after valaciclovir administration is graphically depicted in the simulated concentration-time profiles, generated by superposition of mean concentration data, in subjects with normal renal function (Fig. 3). Administration of high-dose oral acyclovir (800 mg every 4 hours, five times daily) typically yields maximum plasma acyclovir concentrations of approximately 1.6 μg/ml at multiple-dose steady state. This value is less than

$t_{1/2}$ (hr)	
Day 1	*Day 11*
2.66 ± 0.32	3.24 ± 0.75
2.92 ± 0.52	3.30 ± 0.58
2.54 ± 0.34	3.06 ± 0.58*
2.42 ± 0.16	2.90 ± 0.38*
2.69 ± 0.21	3.18 ± 0.19*

that expected from even the lowest dose (250 mg) of valaciclovir used in the multiple-dose study and significantly lower than the average peak levels of approximately 6 and 9 μg/ml expected from the administration of 1000 or 2000 mg valaciclovir four times daily. As a result of its low oral bioavailability, the AUC from oral acyclovir administration typically displays significant intersubject variability, as characterized by coefficients of variation of approximately 25% to 35%.[2] This is greater than that observed after valaciclovir administration at all but the highest doses administered (Tables II and III). Thus, not only does valaciclovir administration result in higher plasma levels, but their variability, at least in normal volunteers, is less than or equal to that from oral acyclovir administration.

The enhanced acyclovir exposure after valaciclovir administration is further characterized in Table VII, in which comparisons of peak concentrations and estimated daily AUC with both oral and intravenously administered acyclovir are presented. Compared with high dose (800 mg) oral acyclovir administration, the relative bioavailability of acyclovir from administration of valaciclovir ranges from approximately 3.3 to 5.5 times greater. In making such evaluations, it needs to be noted that acyclovir accounts for only approximately 69.4% of valaciclovir by weight. Accordingly, for example, complete absorption and conversion of 1000 mg valaciclovir is equivalent to 694 mg acyclovir. At the 2000 mg valaciclovir dose, the peak acyclovir concentration approaches that attainable with 5 mg/kg intravenous acyclovir administered every 8 hours. Estimated daily acyclovir AUC values from 1000 and 2000 mg valaciclovir given four times daily exceed or approximate, respectively, those obtained from 5 mg/kg and 10 mg/kg intravenous acyclovir administered every 8 hours.

Plasma concentrations of valaciclovir at all dose levels were consistently below the lower limit of assay quantitation within 3 hours after administration. This

Table VI. Adverse experiences reported during multiple-dose administration of valaciclovir or placebo

Adverse experience	*Valaciclovir recipients (n = 40)*	*Placebo recipients (n = 20)*
Headache	13	9
Rhinitis	9	5
Pharyngitis	7	1
Pruritus	4	0
Abdominal pain	3	1
Back pain	3	0
Nausea	3	1
Dizziness	3	4
Asthenia	2	0
Diarrhea	2	0
Somnolence	2	0
Cough	2	1

One report of each of the following for valaciclovir recipients: infection, pain, arrhythmia, tachycardia, vasodilation, dyspepsia, flatulence, thirst, vomiting, arthralgia, leg cramps, myalgia, hyperkinesia, nervousness, voice alteration, rash, ear disorder, lacrimation disorder, ear pain, eye pain, and photophobia

Two reports of rash and one report of each of the following for placebo recipients: fever, hypertension, dry mouth, tooth disorder, sweating and dry eyes

suggests that the elimination $t_{1/2}$ of valaciclovir is very likely less than 30 minutes. Estimates of acyclovir $t_{1/2}$ were approximately 2½ to 3 hours and consistent with historical reports.[14,15] The longer $t_{1/2}$ for acyclovir and the high acyclovir/valaciclovir plasma concentration ratios indicate that the disposition of acyclovir after valaciclovir administration is not limited by its formation rate.[16] This fact is important in that it will permit guidelines for dosage adjustment to be made on the basis of estimates of renal function, just as with acyclovir itself.

The less than proportional increases in acyclovir C_{max} and AUC observed after valaciclovir administration are very likely a result of reduced absorption of valaciclovir with increasing dose, rather than saturable conversion of valaciclovir to acyclovir. This is supported by the reduction in total urinary recovery (valaciclovir and metabolites) seen in the single-dose cohort and by the observation of a consistent valaciclovir/acyclovir peak concentration ratio (4%) across the doses investigated. Saturable conversion would be characterized by a dose-related increase in this ratio with increasing dose of valaciclovir. This, in fact, was

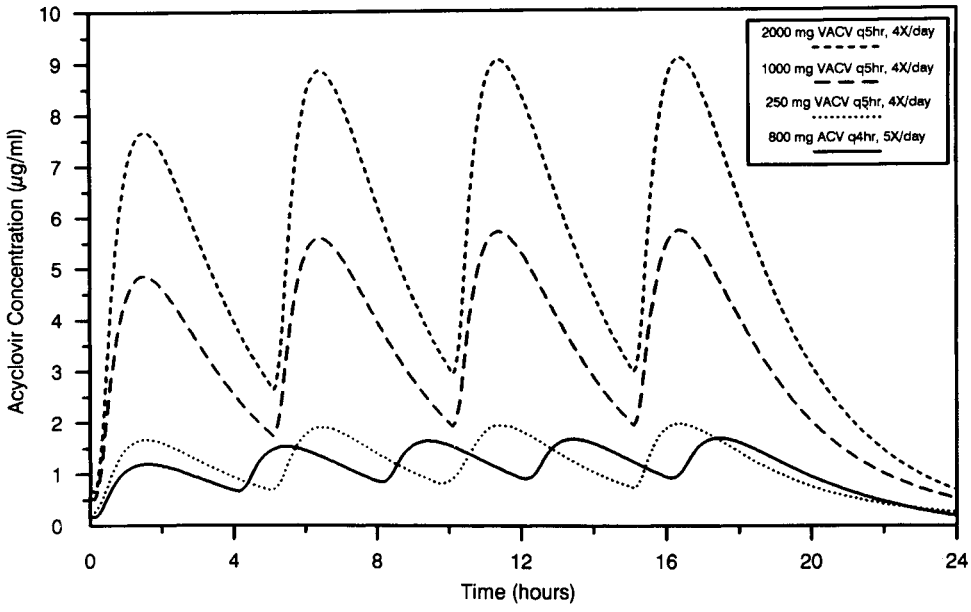


Fig. 3. Simulated acyclovir (ACV) plasma concentration–time profiles at multiple-dose steady state from administration of 250, 1000, or 2000 mg valaciclovir (VACV) every 5 hours, four times daily, and high-dose oral acyclovir given every 4 hours, five times daily.

Table VII. Comparison of acyclovir exposure from valaciclovir and intravenous (IV) and oral (PO) acyclovir administration

Oral valaciclovir	*Acyclovir*	C_{max} *(μg/ml)*	*Daily AUC (hr · μg/ml)*
—	200 mg 5×/day PO	0.8	12
—	800 mg 5×/day PO	1.6	24
250 mg 4×/day	—	2.1	23
500 mg 4×/day	—	3.7	40
1000 mg 4×/day	—	5.0	63
1500 mg 4×/day	—	6.4	86
2000 mg 4×/day	—	8.5	109
—	5 mg/kg q8hr IV	9.8	54
—	10 mg/kg q8hr IV	20.7	107

Parameter estimates for PO acyclovir from the package insert for Zovirax capsules, tablets, and suspension, Burroughs Wellcome Co., 1991.
Parameter estimates for IV acyclovir from Blum et al.[14]
Parameter estimates after valaciclovir administration from multiple-dose study.

not observed. When viewed in conjunction with an observed increase in acyclovir t_{max} with increasing valaciclovir dose, the existence of a dose-related decrease in valaciclovir bioavailability suggests that absorption sites for valaciclovir may be proximally saturated along the gastrointestinal tract. However, there was no evidence from the multiple-dose study that repeated administration of valaciclovir for 11 days results in any appreciable reduction in acyclovir concentrations or drug exposure over time.

The findings from the placebo-controlled single- and multiple-dose studies indicate that administration of valaciclovir is well tolerated at oral doses ranging from 100 to 2000 mg administered up to four times daily for 11 days. In the single-dose study, there were no appreciable differences between drug and placebo recipients in the nature or frequency of adverse events judged to be possibly associated with study medication. Adverse events observed in the multiple-dose study were consistent with those observed for acyclovir. Acyclovir has a large therapeutic index as indicated by the fact that, in conjunction with its low oral

bioavailability, current labeling recommends oral dosage reductions only in cases of moderately severe to severe renal impairment. Rarely, serious central nervous system adverse events and obstructive nephropathy have been associated with acyclovir administration. These have occurred predominantly in immunocompromised patients receiving intravenous acyclovir or in patients with renal dysfunction receiving higher than recommended doses or who have inadequate hydration.[1] Although it has been difficult to specifically attribute these events to acyclovir (they may often be attributable to other medications or to underlying disease), given the enhanced bioavailability of acyclovir from valaciclovir administration, it can be anticipated that dosage adjustments in addition to those for oral acyclovir may be needed to maintain acceptable risk/benefit ratios in patients with impaired renal function. It is noteworthy that no renal dysfunction or crystalluria occurred in these valaciclovir studies. The acyclovir AUC values resulting from high-dose valaciclovir administration are similar to those historically observed with high-dose intravenous acyclovir, but the peak concentrations are appreciably less. This suggests that the renal adverse events occasionally observed with intravenous acyclovir may be attributable to high peak levels rather than AUC and that such adverse events may well be minimized with administration of valaciclovir.

In summary, valaciclovir, the L-valyl ester of acyclovir, is a pro-drug that displays enhanced bioavailability and significantly higher plasma concentrations of acyclovir than from oral administration of acyclovir itself. On the basis of preclinical toxicology and the safety data from the studies presented here in normal volunteers, valaciclovir has been shown to have the same safety profile as acyclovir. Valaciclovir is currently being evaluated in additional phase I trials in subjects with renal or hepatic impairment and in phase II/III clinical trials of herpes group infections including treatment of zoster, treatment and suppression of HSV, and suppression of cytomegalovirus in immunocompromised patients.

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